



RD-Agent: LLM-based autonomous evolving agents for industrial data-driven R&D

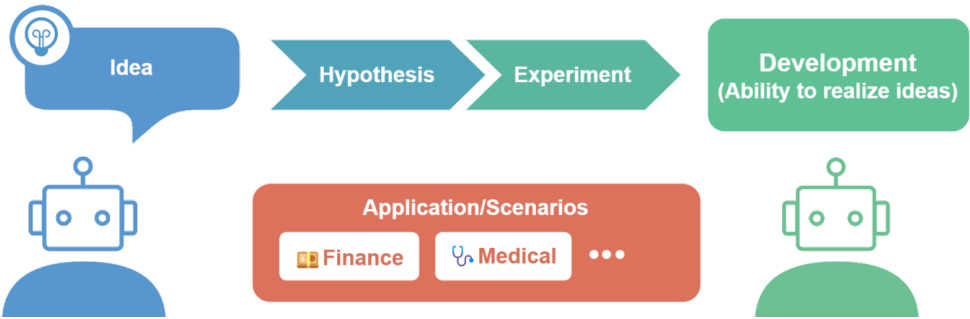
Scenario Description

R&D Agent-Qlib: Automated Quantitative
Trading & Iterative Factors
Evolution Demo

Overview

The demo showcases the iterative process of hypothesis generation, knowledge construction, and decision-making. It highlights how financial factors evolve through continuous feedback and refinement.

Automated R&D





- R (Research)
 - Iterative development of ideas and hypotheses.
 - Continuous learning and knowledge construction.
- D (Development)
 - Progressive implementation and code generation of factors.
 - Automated testing and validation of financial factors.

Objective

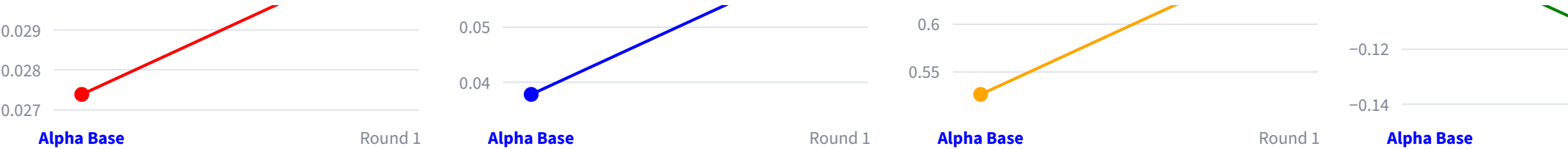
To demonstrate the dynamic evolution of financial factors through the Qlib platform, emphasizing how each iteration enhances the accuracy and reliability of the resulting financial factors.

Summary

Metrics

☐ successful hypotheses





download the metrics (csv)

Hypotheses 🏆

	RD-Agent proposes the hypothesis📄
1	Simple price-volume based factors such as 5-day momentum, 5-day volatility, and 5-day volume change can capture short-term trends and risk, potentially predicting future returns.

☒ show analyse_task_competition

Task Completion Analysis

🔄 Aggregate Completion Across All Loops

Loop	Total Tasks	Passed (Round 1)	Passed (Round 3)	Passed (Round 5)	Passed (Final)
Loop 1	3	2/3 (67%)	2/3 (67%)	3/3 (100%)	3/3 (100%)

TOTAL	3	2/3 (67%)	2/3 (67%)	3/3 (100%)	3/3 (100%)
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Overall Completion Progress:

After Round 1 	After Round 3 	After Round 5 	Final Completion 
67%	67%	100%	100%
	 0%	 33%	 0%

Detailed Results by Loop

 Loop 1 Details
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R&D Loops

Loops

 1

Research 	Development 
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Hypothesis💡

- **Hypothesis:** Simple price-volume based factors such as 5-day momentum, 5-day volatility, and 5-day volume change can capture short-term trends and risk, potentially predicting future returns.
- **Reason:** These factors are simple, easy to compute, and have economic rationale: momentum captures trend persistence, volatility captures risk premium, and volume change captures liquidity and trading activity. They use only price and volume data, making them robust and interpretable. Starting with these will establish a baseline for further refinement.

Factor Tasks▶️

5_day_momentum 5_day_volatility 5_day_volume_change

Description: [Momentum Factor] 5-day momentum captures short-term price trend persistence. It is calculated as the ratio of the current close price to the close price 5 days ago minus 1.

Formulation

$$\text{Momentum}_t = \frac{P_t}{P_{t-5}} - 1$$

Variable	Description

📌 Evolving Status

1	2	3	4
✓	✓	✓	✓
✓	✓	✓	✓
✗	✗	✗	✓

🔄 Evolving Rounds

☐ 1 ☐ 2 ☐ 3 ☒ 4

5_day_momentum✓ 5_day_volatility✓ 5_day_volume_change✓

Workspace Path: /workspaces/rdagent-qlib-cloud/git_ignore_folder/RD-Agent_workspace/310a3ed2eb284f7e80520e0ba783e8ec

▼ factor.py

```
import pandas as pd
import numpy as np

def calculate_5_day_momentum():
    # Read the daily price and volume data
```

P_t	Close price at day t
P_{t-5}	Close price at day t-5

Feedback

workspace

/workspaces/rdagent-qlib-cloud/git_ignore_folder/RD-Agent_workspace/c7e8dc5294ff44cca6493bf503168de1

▼ Config

Dataset 	Model 	Factors 	Data Split 
CSI300	LGBModel	Alpha158 Plus	Train: 2008-01-01 to 2014-12-31 Valid: 2015-01-01 to 2016-12-31

```
df = pd.read_hdf('daily_pv.h5', key='data')

# Sort the dataframe by datetime and instrument
df = df.sort_index(level=['datetime', 'instrument'])

# Calculate the 5-day momentum factor
# Group by instrument and shift the close price
df['close_5d_ago'] = df.groupby(level='instrument')['close'].shift(5)
df['5_day_momentum'] = df['$close'] / df['close_5d_ago']

# Select only the datetime, instrument, and factor
result = df[['5_day_momentum']].dropna()

# Save the result to result.h5
result.to_hdf('result.h5', key='data', mode='w')

return result

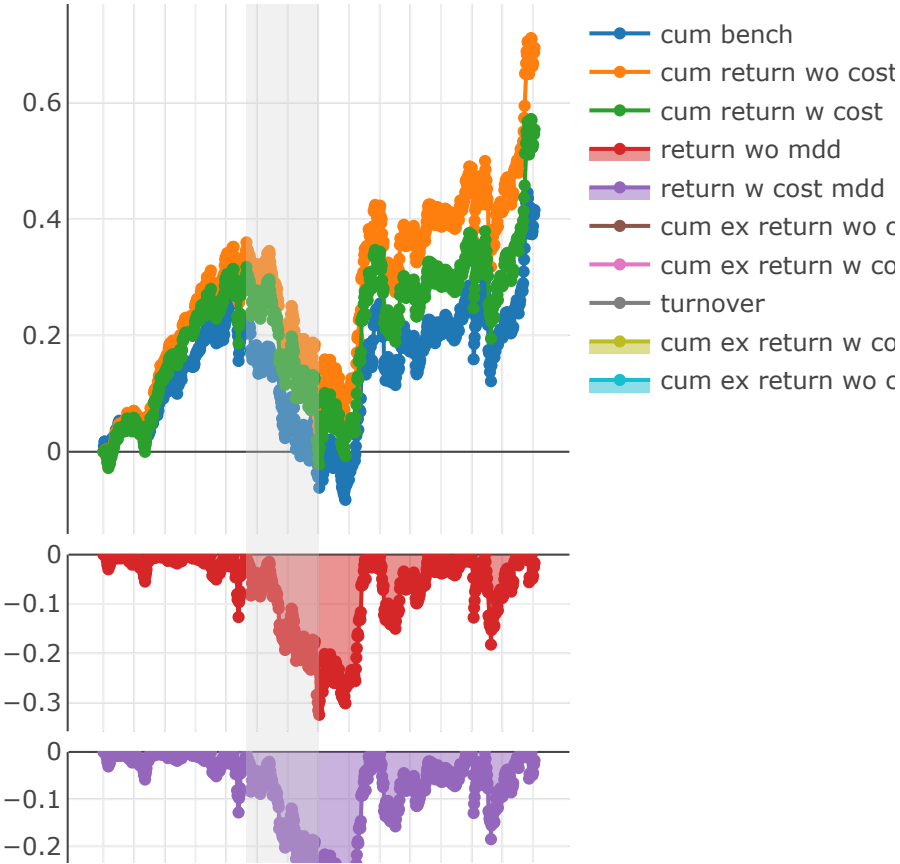
if __name__ == '__main__':
    calculate_5_day_momentum()
```

Final Feedback Execution Feedback Code Feedback Value Feedback

value feedback: value feedback: The source dataframe has only one column which is correct. The source dataframe does not have any infinite values. The output format is correct: it is a pandas DataFrame with a MultiIndex (datetime, instrument) and a single column named '5_day_momentum' with float32 dtype. The non-null count matches the

		Plus	2010-12-31
			Test : 2017-01-01 to
			2020-08-01

Returns

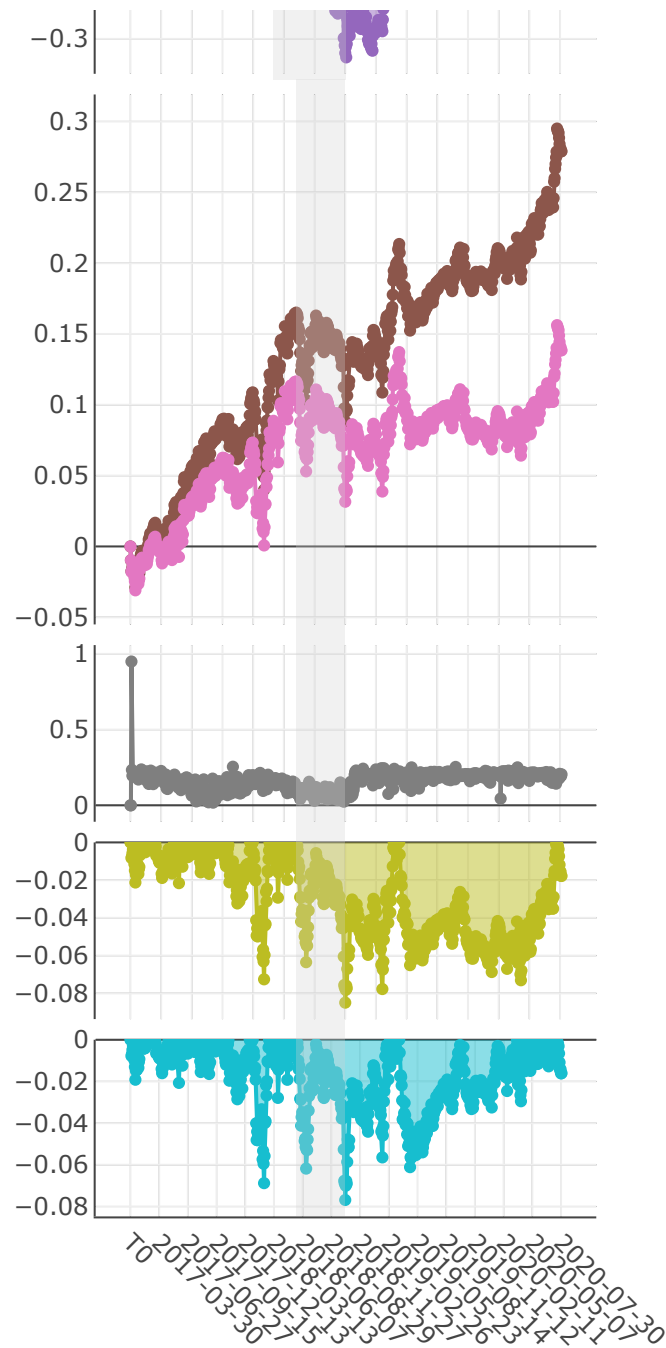


total entries, which is acceptable. The generated dataframe is daily.

shape feedback: None

shape feedback: value feedback: The source dataframe has only one column which is correct. The source dataframe does not have any infinite values. The output format is correct: it is a pandas DataFrame with a MultiIndex (datetime, instrument) and a single column named '5_day_momentum' with float32 dtype. The non-null count matches the total entries, which is acceptable. The generated dataframe is daily.

shape feedback: None



Hypothesis Feedback🔍

- **Observations:** The combined factors (5-day momentum, 5-day volatility, and 5-day volume change) yield an IC of 0.030954, which is higher than the SOTA IC of 0.027387. The annualized return with cost is 0.063456, significantly higher than the SOTA's 0.037827. However, the max drawdown is -0.136414, which is worse (more negative) than the SOTA's -0.084930.
- **Hypothesis Evaluation:** The hypothesis that simple price-volume based factors can capture short-term trends and risk is supported. The improvement in IC and annualized return indicates that these factors provide predictive power for future returns. The higher drawdown suggests increased risk, but the overall return improvement is substantial.
- **New Hypothesis:** Incorporating additional short-term factors such as 5-day average true range (ATR) or 5-day relative strength index (RSI) could further enhance predictive power while potentially managing drawdowns. Additionally, combining these factors with a risk overlay or volatility scaling might improve risk-adjusted returns.
- **Decision:** True
- **Reason:** The current factors show strong predictive ability (higher IC) and return generation, but the increased drawdown indicates a need for risk management. Adding complementary technical indicators or adjusting factor weights could help balance return and risk. Since the current factors are already in the SOTA library, exploring new

directions like ATR or RSI could provide additional alpha and diversification.

Disclaimer

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